

Nitrogen use and losses in agriculture in subtropical Australia

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Abstract

This review examines the use of nitrogen (N) fertilizer on sugar-cane, summer and winter grain crops, cotton, tropical fruit crops and pastoral areas in the four subtropical zones in eastern Australia. The pathways for N loss from the various crops grown in these zones are also examined and estimates of N loss given.

Sugar-cane is the most important crop grown in the subtropical humid northern and southern zones, using 77% of all N fertilizer applied in 1988–89. Urea is the most widely used form of N fertilizer with about 50% of the applied N often lost via ammonia volatilization, denitrification and leaching. Losses of N via ammonia volatilization can be reduced by either irrigating after application, applying urea in subsurface bands or delaying application until after canopy development. Denitrification losses of 20% of applied N have been measured on clay soils in sugar-cane areas while leaching losses may occur by movement of solutes down preferential pathways (e.g. soil fauna, root channels and structural weaknesses in the soil profile). Tropical fruit crops also make a significant contribution to the economy of the humid northern and southern zones. The livestock industry is well established in the subtropical northern zones, with beef and dairy production relying on leguminous as well as N fertilized pastures. Urea is again the most widely used form of N and is susceptible to large losses via ammonia volatilization. Over a 12 month period, losses of between 9% and 42% of the N applied were recorded from a subtropical pasture.

Wheat is the major winter crop of the sub-humid northern and southern zones with grain sorghum the main summer crop. Urea is the principal form of N fertilizer applied to both crops and is essential for increasing or maintaining economic yields from both regions. This decrease in soil fertility in grain producing areas is due mainly to a decrease in the amount of soil organic matter available for mineralization. Cotton is another major crop of both areas and relies heavily on N fertilizer application. Nitrogen fertilizer losses have been recorded from all cropping areas, although nitrification inhibitors such as wax coated calcium carbide and 2-ethynylpyridine have reduced denitrification losses from soils growing wheat and cotton respectively.

Subtropical agriculture relies heavily on N fertilizer, principally urea, to maintain and increase crop yields. Losses of N from soils sown to crops and from native and sown pasture occur although management practices are being developed to help minimize this loss.

Introduction

The subtropical regions of the world include areas in the northern hemisphere in India, Indo-China, southern China, Mexico and Florida (USA) while, in the southern hemisphere, the subtropics cover areas of southern Africa, Madagascar, South America and areas of Australia along the eastern and western coastlines. In

80% of this land, rainfall is the limiting factor for crop production with further limitations imposed by soil, topography and farming practices. Technologies developed for temperate regions were not always suitable for subtropical climates with the result that plants and animals were bred or selected for the subtropical regions from highly developed research programs. Even though agricultural land in subtropical regions

Table 1. Typical crops grown in subtropical Australia, the average yield, the quantity of fertilizer applied per hectare per year and the amount removed in each crop

Crop	Yield ¹ (t ha ⁻¹)	Fertiliser applied (kg N ha ⁻¹ yr ⁻¹)	N removed in crop ² (kg N ha ⁻¹)
Barley	1.84	90	33
Grain sorghum	1.68	50	24
Maize	3.61	135–225	54
Oats	1.01	100	18
Wheat	1.66	30–60	33
Sugarcane	82.80	160–200	64
Cotton	2.75	120	65
Bananas	29.72	225	48
Pineapples	41.69	300	67

¹Australian Bureau of Statistics (1994) Crops and Pastures Queensland 1992–93 Catalogue No. 7321.3;

Australian Bureau of Statistics (1994) Fruit Queensland 1992–93 Catalogue No.7322.3;

Australian Bureau of Statistics (1989) Crops and Pastures 1987–88 New South Wales Catalogue No.7321.1.

²Vallis (1990) [58]

throughout the world is diverse, cropping and nitrogen (N) fertilization practices in the different regions are often quite similar. Thus, results from research conducted in one geographic region are generally applicable to other regions within the subtropical zones.

This review covers N research in subtropical regions of eastern Australia. These regions can be divided into four geographically and climatically distinct zones between latitudes 20°S and 34°S[6]:- (1) Subtropical, humid north (2) Subtropical, humid south (3) Subtropical, sub-humid, north and (4) Subtropical, sub-humid, south (Fig. 1). The total area of land covered by these four zones is approximately 57.4 M ha which is 22.7% of the combined area of Queensland and New South Wales but only 7.5% of Australia's total area.

The major crops grown in these regions include sugar-cane, wheat, grain sorghum, cotton, soybeans, sunflowers, oats and barley while tropical fruits form a significant proportion of agricultural produce from the two coastal regions. Of these crops, pineapples require the highest rates of fertilizer N, with grain sorghum and wheat the least users of N (Table 1) [2,4,5]. Average values for N removed by the crop ranged from 18 kg N ha⁻¹ for oats to 67 kg N ha⁻¹ for pineapples [58]. This N 'loss' is in addition to losses of N through biological processes.

The sub-humid, northern zone carries about 10% of Australia's total beef cattle herd while the sub-humid southern zone supports both sheep and cattle herds. In this review, the use of N fertilizer on crops and pastures in these four zones will be discussed in more detail together with the possible mechanisms for N loss and their relative importance for different crops.

Zones 1 and 2 - The subtropical humid northern and southern regions

In these two subtropical, humid zones, with areas of 1.249 and 2.009 M ha respectively, annual rainfall ranges from 1672 mm to 1200 mm in the northern zone, while in the south, average rainfall varies from 2000 mm at the coast to 900 mm at the western boundary of the region. Maximum summer temperatures average 31°C in December and January in the northern zone and 30°C in the southern zone. Both regions can expect winter frosts. Crop and livestock production in both areas is similar, with sugar-cane the principal crop.

Cropping

Sugar-cane

Fertilizer use

Sugar-cane is grown extensively in both regions, with 40% of Australia's total crop produced in the humid north and approximately 5% in the south. The gross value of this crop in 1991–92 was estimated to be \$1 billion. In the northern region, where sugar-cane is a 1-year crop, green-cane harvesting is now practised on 90% of farms, with about 15 tonnes ha⁻¹ of biomass returned to the soil as a trash blanket. Prior to this change in management practice, much of the biomass other than stem was removed by burning. In the southern zone, where the sugar-cane crop takes 2 years to reach maturity, burning is still practiced. Temperatures are too low for decomposition of the trash blanket to occur over the growing season. As a result, soil moisture is retained for longer periods and some decrease in yield may be experienced.

Most crops of sugar-cane receive 160–200 kg N ha⁻¹ with some irrigated crops receiving in excess of 300 kg N ha⁻¹. In 1988–89, 77% of all N fertilizer used in the northern and southern zones was applied to sugar-cane [48], urea being the most widely used form

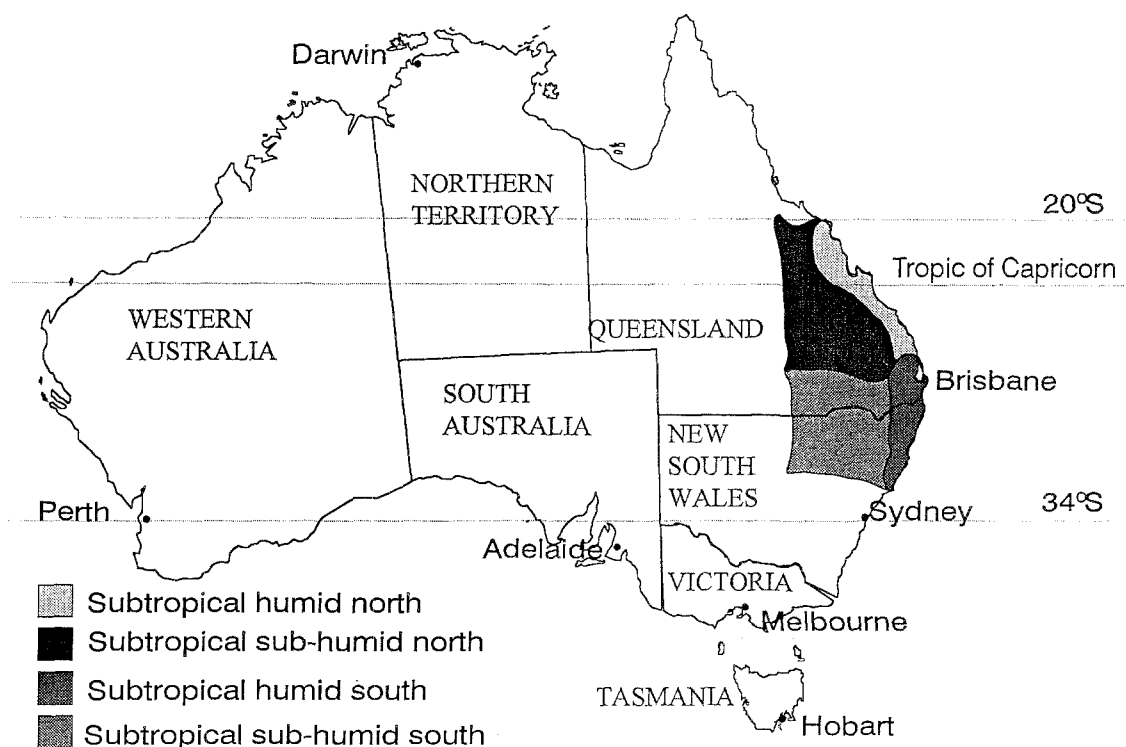


Fig. 1. The sub-tropical regions of Australia can be categorised as four distinct zones:- (1) subtropical, humid north (2) subtropical, humid south (3) subtropical, subhumid, north and (4) subtropical, sub-humid, south. Adapted from Australian Farming Systems, Australian Government Publishing Service, Canberra, 1982.

[3] and ammonium sulphate accounting for 7%. The sugar industry is therefore a major user of N fertilizer with some 60,000 tonnes applied annually.

Fertilizer N loss

Loss of fertilizer N from the sugar-cane crop-soil system ranges from 25% to 60% (Vallis, pers. comm.). This amounts to an annual loss of 15,000 to 36,000 tonnes of N valued at approximately \$15M to \$36M. In an ^{15}N trial in northern New South Wales and southern Queensland, only one quarter of the N applied was recovered in the above-ground biomass after six months [38]. A similar amount was found in the soil and the roots. Ammonia volatilization, denitrification and leaching all seem to be involved in this loss of 50% of the applied fertilizer N with their relative contribution depending upon soil type, weather and management practice.

Ammonia losses of fertilizer N have been accentuated by the change in crop management from burnt-cane to green-cane harvesting. Surface applications of urea resulted in prills being trapped on the trash, making them susceptible to loss through ammonia

volatilization. The pattern, rate and extent of ammonia losses are controlled by the availability of water in the trash and its evaporation [24, 26]. Small amounts of water added to trash resulted in losses of 32% and 39% of the applied N at different sites in north Queensland. Heavy rainfall decreased this loss to 17% while little ammonia was lost when conditions were dry. Substitution of ammonium sulphate for urea has reduced the ammonia loss to less than 1.8% of the applied N [26]. Other avenues to reduce ammonia loss from trash include irrigation after urea application and application of urea when the crop has developed a substantial canopy [21].

Biological denitrification losses of applied N from sugar-cane soils are difficult to determine in the field. The use of microplots and ^{15}N for short-term measurements with extrapolation to an area basis is common practice in denitrification research as the costs of using ^{15}N in combination with micrometeorological techniques is prohibitive. The use of ^{15}N is necessary to gather essential data on the proportion of gases entering the atmosphere as N_2 and N_2O . The potential for denitrification to occur in sugar-cane soils in Queens-

land and northern New South Wales was measured using $^{15}\text{N-NO}_3^-$ in microplots [60]. Losses of $< 1 \text{ kg N ha}^{-1} \text{ week}^{-1}$ were recorded on well-drained red-earth soils in Queensland. On heavy clay soils in northern New South Wales, rates ranging from 2.1 to 12 $\text{kg N ha}^{-1} \text{ week}^{-1}$ were found under zero tillage. This amounted to losses of between 2.1% and 19.7% of the N applied. Soil texture and soil moisture are major factors influencing denitrification rates in the field [62]. A reduction in the amount of N lost through denitrification may be achieved by the use of nitrification inhibitors such as 2-ethynylpyridine which has proven successful in decreasing losses of applied N from a cotton crop [27].

Leaching losses of N from sugar-cane soils were recently assessed using bromide as a tracer of nitrate movement in the soil profile (V.R. Catchpoole pers. comm.). On the well-drained red-earth soils, 30% and 60% of the bromide applied had been lost after 1 month and 3 months respectively. The failure to recover all of the bromide applied to the plots meant that the extent of leaching of anions remained uncertain. However, bromide was lost from all sites with the initial rapid loss possibly due to movement of solutes down preferential pathways (e.g. root channels and soil structural weaknesses). Further evidence for preferential flow was obtained when bromide was banded into a well-drained alluvial soil in the southern humid zone in New South Wales (Table 2). After 6 weeks, 43% of the bromide was unaccounted for with the remainder moving to one side of the application area and down to a depth of 150 cm. The potential for leaching of nitrate-N derived from fertilizers persists in the soil for 6 weeks after application.

Soil erosion is a highly variable and often significant source of N loss from cultivated cropping systems ranging from 1 to 100 $\text{kg N ha}^{-1} \text{ yr}^{-1}$ [50] and seriously affects $> 80\%$ of the cropping lands in Queensland [9]. More than 90% of this N loss occurs as organic matter in sediments and suspensions. Erosion is a major cause of long-term fertility decline in soils under 'conventional' clean cultivation with estimates of total-N losses increasing from 77 $\text{kg N ha}^{-1} \text{ yr}^{-1}$ in the humid southern zone to 220 $\text{kg N ha}^{-1} \text{ yr}^{-1}$ at Mackay in the humid northern zone [50]. This increase is associated with a higher intensity of rainfall in the northern zone which indicates special attention for erosion control in coastal cropping lands of the subtropics. For example, a loss of 1 mm depth of soil corresponds to an annual soil loss of 10 t ha^{-1} containing 20 kg N ha^{-1} . In sugar-cane areas, this equates to a loss of 150–450 kg N ha^{-1}

yr^{-1} [58] and a possible yield reduction of 20% for sites with a shallow topsoil [50]. If zero tillage is adopted as a management practice, soil erosion is reduced and this could be attributed to the undisturbed and compacted nature of the soil surface [47].

Other crops

Tropical fruit crops such as bananas, papaws, mangoes, pineapples, macadamia nuts and avocados are also grown extensively in both northern and southern humid zones. Some 11,000 ha of wheat, oats and barley are grown near the coast in northern New South Wales, but because of similarities in production with the inland, sub-humid zone, N use on these crops will be reviewed later in this article. Coarse grain crops such as grain sorghum and maize together with soybeans are grown in both zones, but do not constitute major crops of either region and hence their cropping practices will not be discussed. Information is available on N use on these crops and on annual removal of N in the harvested product, but there are few, if any, studies of other N loss processes.

Bananas

Banana production is an important and valuable industry in Australia with a gross value in 1990/91 of \$178M. The national average yield per hectare is 20 tonnes with supply dominated by the Queensland coastal belt [1]. Nitrogen fertilizer applications are important because the crop needs more N to cover the large losses of N than can be supplied by the soil and through irrigation. The recommended application rate for the crop is 225 kg N ha^{-1} with split applications recommended to minimize leaching losses [64]. Urea is the main N fertilizer in use, both for soil application and as a foliar spray to the leaves. A harvested crop of 20 tonnes ha^{-1} removes 47 kg N ha^{-1} , the majority of the plant N being in the pseudostem which is returned to the soil.

Pineapples

The pineapple industry is based in south-east Queensland in the humid, southern zone where approximately 80% of the crop is grown. The average yield is 38 tonnes ha^{-1} with a gross value of \$49M in 1990/91 [1]. Heavy fertilization of the crop is required for continued production, with N being applied at a rate of 300 kg ha^{-1} in mixtures of phosphorus and potassium fertilizers at 3 and 6 months. This is supplemented

Table 2. Concentration of bromide in the soil profile ($\text{kg Br}^- \text{ ha}^{-1}$) after 6 weeks following the application of $200 \text{ kg Br}^- \text{ ha}^{-1}$ to a sugar-cane crop at Harwood during the 1991/92 season (Catchpoole, unpublished)

Time weeks	Depth (cm)	Position in Plot*					Total
		4	1	2	3	5	
0	0–10	0		198.2		0	198.2
2	0–10	0		180.9		0	180.9
4	0–10	1.3		49.2		0	50.5
	10–20	0	8.0	19.1	29.0	0	56.1
	20–30	0	2.3	3.8	5.5	15.1	26.7
	30–40	0	0	0	0.9	6.6	7.5
	40–50	0	0	0	0	5.0	5.0
	Total						145.8
6	0–10	0		7.2		0	7.2
	10–20	0	3.5	3.3	2.3	0	9.1
	20–30	0	3.6	8.7	5.5	0	17.8
	30–40	0	0	7.1	3.4	4.0	14.5
	40–50	0	1.8	3.8	2.1	0	7.7
	50–60	0	0	0	2.4	1.9	4.3
	60–80	0	0	0	3.7	5.2	8.9
	80–100	0	0	0	3.9	10.9	14.8
	100–120	0	0	0	3.6	9.3	16.7
	120–150	0	0	0	2.7	10.6	13.3
	Total						114.3

*The numbers indicate the positions on either side of the cane row, with 2 the middle of the row, 1 and 3 20cm on each side of 2, and 4 and 5 40cm on each side of 2.

with about 90 kg N, as ammonium sulphate, at 1.5 and 4.5 months. Fertilizer application is generally to the soil or leaf axils although the use of urea as a foliar spray is increasing. Loss of N through leaching does not appear to be a problem although some leaching must occur following significant rainfall events. Irrigation is not a common management practice as the plant has the ability to store large volumes of water. Ammonia volatilization and denitrification are not thought to be important avenues for loss of N from pineapple fields.

Avocados

The production of avocado fruit in Australia in 1989 was 13,000 tonnes, up from 662 tonnes in 1977. The gross value of the crop in 1991–92 was \$25.4M with Queensland producing 60% of the crop and the remainder produced in the humid southern zone of New South Wales. Inorganic fertilizers are used to supplement

organic nutrient applications with 3 split doses between summer and mid-autumn. Fertilizer application rates are usually based on soil and leaf analysis as well as on tree size. Critical leaf N levels for avocado have been established [46] with 1.6%–2.0% N considered adequate for growth.

Mangoes

Mango production in Australia occurs in the sub-tropical humid northern and southern zones with the Bowen-Burdekin region in the north the major producing area. In 1986/87, 17,000 tonnes of mangoes was produced with a value of \$45M. The nutrient requirements of the crop increases when trees start bearing (1–5 years). For young trees, 4 equal split applications of high N fertilizer are required per tree per year. For bearing trees, 3 to 5 kg N per tree per year is required with 50–75% being supplied in December and 25–

50% in February. Foliar applications of N are also needed.

Macadamia nuts

Macadamia is a native of the Australian subtropical east coast rain forests. The trees are now grown commercially in orchards with the macadamia nuts harvested for export. The crop is grown extensively in northern New South Wales and southern Queensland with about 4,500 tonnes of nuts produced in 1990/91 with a value of \$18M. The fertilizer requirement is 120 g N tree⁻¹ year⁻¹ of age and this is split into 2 or 3 applications.

Pastoral

The productivity of the beef cattle industry in the humid northern and southern zones is directly related to the use of either improved legume-based pastures or N-fertilized grass pastures. In the northern zone, legumes such as stylo (*Stylosanthes humilis*) and siratro (*Macropitilium atropurpureum*) contribute large amounts of N to the soil/plant system, but estimates of their total contribution is difficult to obtain because of the uncertainty of the area and productivity of leguminous pastures [58]. The area of N-fertilized native and sown pastures in this region in 1988/89 was estimated to be 29,740 ha to which 1940 tonnes of N was applied, with urea comprising 89% of the total application [3]. In the northern zone, it was estimated that the live weight gain per ha of beef cattle at optimum stocking rates in N-fertilized subtropical pastures was 1.9 times that of legume-based pastures [37]. In the southern zone, 10,980 tonnes of N fertilizer was applied to 77,730 ha of native or sown pasture [3]. Urea was again the most widely used N source accounting for 87% of the N applied. Beef cattle production on the coastal lowlands, north of Brisbane and on the north coast of New South Wales is based on N fertilized grass pastures. The dairy industry is however the largest user of N fertilizer accounting for 80% of all N usage on pastures [44]. Increased milk production was obtained from pastures fertilized with 400 kg N ha⁻¹ yr⁻¹ [18] while, more recently, application rates of up to 500 kg N ha⁻¹ yr⁻¹ to irrigated pastures [14] has led to increased farm efficiency on dairies in the subtropics [41].

Loss of N from pastures

The major pathways for loss of N from fertilized pastures in these two zones include ammonia volatilization from animal excreta and fertilizers and leaching. Losses of N can also occur via biological denitrification and erosion although, compared with losses measured under crops, they are not seen as significant pathways of loss.

In a study on N losses from urine patches in subtropical south-east Queensland [56], ammonia volatilization accounted for 18.8%, 14.4% and 28.4% of the N applied in urine in winter, summer and autumn respectively. More than half the loss occurred within 48 h of urine application. In a similar study in subtropical north Queensland [57], the apparent loss of N from cattle urine due to ammonia emission, as measured using ¹⁵N, was 18% in autumn, 32% in winter and 16% in spring. Losses were increased from 32% to 44% in winter if the soil was wet with 50 mm water before urine application. Management of animals also impacts on the N cycle and production of pastures. In set-stocked pastures, volatilization losses may be lower because of the ability of the pasture to better utilize excretal N and because plant cover can reduce ammonia losses [23].

For broad area assessment of ammonia volatilization from pastures, micrometeorological techniques have been developed [20] and are used extensively. They have shown that ammonia losses from urea broadcast on to a subtropical pasture ranged from 9% to 42% of the N applied [11]. These rates of loss were confirmed by a parallel ¹⁵N microplot experiment (Table 3) [31]. Losses are highest when urea remains on the surface and is partly solubilized by either small falls of rain or dew. Suggested practices for minimizing this loss include making applications to dry soil just before expected significant falls of rain or by irrigating well after urea application.

Investigations into leaching losses from subtropical pastures in the humid northern and southern regions have been minimal with no quantitative estimates available on the amount of N lost. Nitrogen is usually applied to the pasture as urea at 300 kg ha⁻¹ yr⁻¹ [54], with application timed to maximize N uptake into the plant and thereby minimize N losses through leaching. When ¹⁵NH₄ ¹⁵NO₃ was applied at 150 kg N ha⁻¹ to a subtropical Rhodes grass (*Chloris gayana*) pasture, a significant amount of ¹⁵N was found at depths below 60 cm suggesting that leaching of N below the root zone had occurred [10]. However, no quantitative esti-

Table 3. Ammonia loss from a tropical pasture as determined by microclimatological and ^{15}N methods (after Catchpoole *et al.*, 1983)

Season	Application date	Measurement period - days after application	Net NH_3 -N loss	^{15}N loss	^{15}N loss/net NH_3 -N loss
			(as % of urea N applied)		
Summer	Feb. 22	0 - 14	12	29	2.4
		15 - 91	9	a	a
Autumn	May 24	0 - 14	42	45	1.1
		15 - 91	6	a	a
Winter	Aug. 23	0 - 14	13	23	1.8
		15 - 91	-8	a	a
Spring	Nov. 8	0 - 14	9	21	2.3
		15 - 91	13	a	a

a. No measurement made.

mate could be put on this loss. Research results from the United Kingdom [17] and Canada [40] indicate that leaching losses are much greater from cultivated land than from pasture. Using OECD data based on a fertilizer input of $100 \text{ kg N ha}^{-1} \text{ yr}^{-1}$ and 250 mm of drainage, leaching losses of only $4 \text{ kg N ha}^{-1} \text{ yr}^{-1}$ could be expected from pastures but up to $23 \text{ kg N ha}^{-1} \text{ yr}^{-1}$ from cultivated soils.

Some reduction in leaching losses is exhibited in soils having a significant anion exchange capacity [28]. These soils (krasnozems, red and yellow earths and podzolics with >20% clay) show a net positive charge at depth due to low pH and low soil organic matter. Thus, the nitrate ions are electrostatically held and leaching of N out of the root zone is reduced.

Denitrification may be a major avenue for N loss from fertilized subtropical pastures, but again quantitative estimates are lacking. Some idea of possible losses may be obtained from results found in the field for a non-fertilized green panic (*Panicum maximum* var. *trichoglume*) pasture using the acetylene inhibition technique. N losses ranged from $9.2 \text{ g N}_2\text{O-N ha}^{-1}$ in summer to $26 \text{ g N}_2\text{O-N ha}^{-1}$ in spring [59], the process being limited at all times by low concentrations of NO_3^- -N in the soil profile. Quantitative estimates for the potential for denitrification to occur in fertilized pastures were obtained from a laboratory incubation study, where maximum gaseous N loss was 1.7% of the N applied. Nitrous oxide constituted 15% of this loss and dinitrogen 85% [61]. This study also confirmed that the seasonal variation in denitrification rates found in the field for green panic pastures could be verified

using laboratory incubation of intact soil cores. The latter technique may thus enable a better estimate of N losses via denitrification to be made over the coming years.

Soil erosion has been estimated to affect 51% of rural land in Australia with the most fertile surface layer removed first [65]. However, these erosion losses of total N are probably occurring quite slowly, taking a number of years before they become visible. In southern central Queensland, topsoil lost from land cleared and grazed with sheep was estimated to be 15 mm over 17 years [49] constituting a possible loss of 150–450 kg N ha^{-1} . However, in general, erosion losses of N from pastures, except where severely overgrazed, are likely to be much lower than from cropping areas.

Zones 3 and 4 - The subtropical, sub-humid, northern and southern zones

In the inland, sub-humid, northern zone (area 24.5 M ha), average annual rainfall is 800 mm and ranges from 1000 mm in the east to 550 mm in the western inland. Sixty to seventy percent of total rainfall is received between November and March (summer) while reasonable rainfall also occurs from May to July (winter) allowing for the planting of winter grains. Evapotranspiration rates are high and range from 76 mm month^{-1} in winter (June/July) to $250 \text{ mm month}^{-1}$ at the beginning of summer (December). Average minimum and maximum temperatures range from 17°C to 32°C in summer to 9°C and 24°C in the cooler months. Crop-

ping and pastoral activities are of equal importance in this zone with grain sorghum, wheat and cotton the principal crops, and cattle the principal livestock. This region carries about 10% of Australia's beef cattle herd.

The sub-humid, southern zone, with an area of 29.6 M ha, can be divided into three climatic regions [6]:- (1) Condamine-Maranoa basin where average yearly rainfall ranges from 650 mm in the east to 450 mm in the west. Average summer temperatures are about 26°C with a fall to a winter average of 12°C. (2) Darling Downs - average summer rainfall ranges from 1000 mm in the east to 650 mm in the west. Air temperatures range from a winter minimum (July) of 3°C to a summer maximum (January) of 30°C. (3) North-West Slopes and Plains and Northern Tablelands of New South Wales. Rainfall and temperatures are both affected by altitude and distance from the ocean. On the Northern Tablelands, average rainfall varies from 736 mm in the east to 550 mm in the west. Temperatures also vary markedly, ranging from a summer maximum in the east of 27°C to 35°C in the west. Winter minimums of 0.4°C and 3.4°C can be expected from east to west. All three regions can expect frosts, but the New South Wales region experiences a greater number of severe frosts. Wheat, barley, oats and safflowers are the principal winter crops while grain sorghum, cotton, sunflowers and soybeans are the more important summer crops. The pastoral industry is also important to this sub-humid, southern zone with sheep, beef cattle and dairying contributing significantly to the income of this area.

Winter cropping

Wheat

Premium high protein hard wheat is the major winter crop in both the northern and southern zones. In 1991–92, the total value of the Australian wheat crop exceeded \$2 billion with the national average yield at 1.5 t ha⁻¹. Production of barley in 1991–92 was valued at \$650M with an average yield of 1.7 t ha⁻¹. In the southern zone, oats is grown primarily for green fodder and safflowers constitute about 1% of the winter crops grown. The principal form of nitrogenous fertilizer used in both zones in 1988–89 was urea with a total of 4000 t applied to 170,000 ha of wheat in the sub-humid northern zone and in the Darling Downs and Condamine-Maranoa regions of the southern zone. On

the North-West Slopes and Plains and Northern Tablelands of the southern zone, where 821,000 ha of wheat yielded 1.74 t ha⁻¹ [2], urea was the preferred nitrogenous fertilizer with 19% of growers [43]. The use of diammonium phosphate and blended forms of N fertilizers is increasing and, in 1988–89, constituted 41% of the total N applied to the above areas.

The adoption of N fertilizer use by the northern wheat farmers has been slow and there has been a gradual decrease in soil fertility over the last 40 years, the change being most rapid during the first 10 years of cultivation [29]. For those growers who, in the past, used N at sowing, the amount applied (30 kg ha⁻¹) was lower than the recommended rates of 30–60 kg N ha⁻¹ and resulted in below maximum yields [43]. More recently, large wheat yield responses to fertilizer N, at application rates of between 80 and 106 kg N ha⁻¹, have been recorded [22]. On the Darling Downs in Queensland, a decrease in soil fertility has occurred due to the long-term effects of cultivation and cropping. Fertilizer N is now needed to maintain yields of grain from wheat. Some soils need more than 200 kg N ha⁻¹ to produce a viable return [52]. For irrigated wheat, splitting the N application into equal quantities applied before sowing as anhydrous ammonia and to the foliage at tillering as urea solution produced similar grain yields to those found when all the fertilizer was applied as anhydrous ammonia at sowing [53].

Pathways of N loss

Ammonia volatilization from urea applied to the soil surface of irrigated wheat was reduced by 25% if irrigation was delayed 6 days after fertilizer applications while increasing the quantity of irrigation by 75% reduced losses from 35 to 14 kg N ha⁻¹ [7]. Partial burial of the urea reduced volatilization losses by 80% while partial burial of ammonium nitrate eliminated volatilization losses. Broadcasting urea onto no-till plots of wheat at crop establishment, where an alkaline ash bed had formed on the soil surface following stubble burning, resulted in 24% of the applied N being lost as NH₃. Retaining the stubble on the plots and not burning reduced the ammonia loss to 21%. Delaying urea application until the development of a plant canopy significantly reduced these N losses and also increased grain yield [8].

Denitrification losses of N from irrigated wheat soils ranged from 1% to 14% of applied N [8]. Application of urea to the growing crop several weeks before irrigation minimized this loss. Denitrification losses

from irrigated wheat have also been reduced by the use of wax coated calcium carbide as a nitrification inhibitor. Nitrate production was effectively inhibited for lengthy periods and N use efficiency in wheat crop improved [25]. On the Darling Downs, low recovery of N from flood irrigated wheat was apparently due to denitrification [53] while N losses from conventional and zero tillage treatments of continuous wheat were related to the quantity of fresh substrate available for denitrification [36].

Leaching of fertilizer N into the subsoil beneath winter cereal crops has also been measured [51]. In a long-term fertilization trial at Hermitage on the Darling Downs, leaching was the major pathway of N loss from the system. Nitrate-N from urea applied to wheat and barley at annual rates of 23 and 69 kg N ha⁻¹ respectively, was found to move beyond 210 cm in the soil profile with a distinct bulge occurring at 60–150 cm in the higher fertilizer treatment (Fig. 2) [55]. Elsewhere, there is also evidence of leaching of surface mineralized N below the rooting depth of oats with 57 kg N ha⁻¹ being lost during two very wet years [12].

Erosion is also an important, and perhaps more widespread, loss mechanism for soils undergoing long periods of cultivation and cropping in the inland zones. N can be lost from a vertisol by erosion at the rate of 0.8% year⁻¹ from the top 15 cm during the first 25 years of cultivation [42]. However, the elimination of stubble burning and the retention and incorporation of stubble into the surface soil is proving to be an effective protective measure against soil loss. There is evidence that this practice will not limit plant growth by immobilizing soil N [63].

Summer crops

Grain sorghum

Grain sorghum is the most important summer cereal grown in Australia with a gross value of \$157.5M in 1991–92. The main areas of production in the sub-humid southern zone are the northern and north-western slopes and plains of New South Wales and the Darling Downs in Queensland. In the northern zone, sorghum is grown mainly on the Central Highlands where 40% of Australia's crop is produced. In 1991–92, the area in these two regions sown to sorghum was 532,000 ha with yields ranging from 1.7 to 2.6 t ha⁻¹ [1].

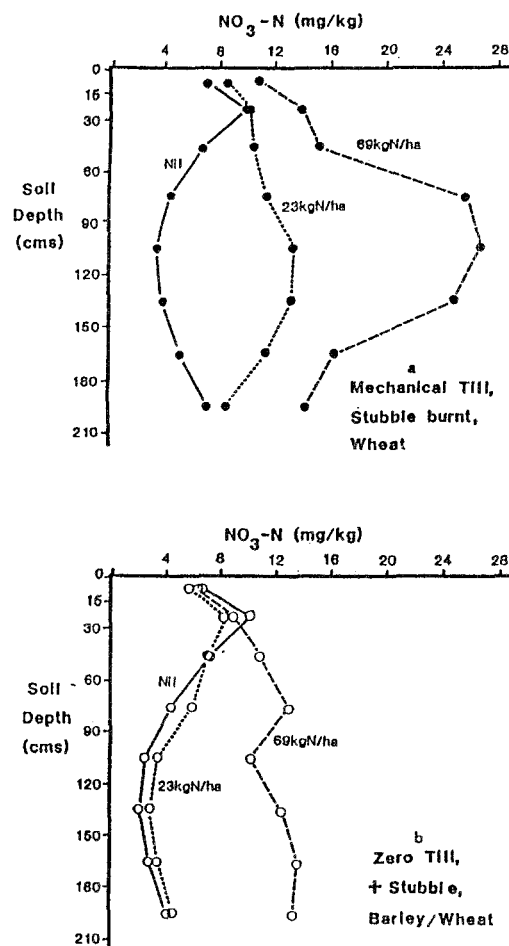


Fig. 2. Distribution of nitrate-N in the soil profile at three rates of N fertiliser applied annually for 18 years. Samples taken in 1986 prior to fertiliser application. (a) treatment combination of mechanical till, stubble burnt and continuous wheat culture. (b) Treatment combination of zero till, stubble retained and wheat/barley culture (after Thompson, 1989).

The use of N fertilizer on grain sorghum production has increased following recognition that decreasing yields are related to decreasing soil fertility. On the Central Highlands in Queensland, where summer-grown crops rely heavily on a good water supply, responses have been obtained by sorghum to both irrigation and N fertilization. When urea was applied at 50 kg N ha⁻¹ under rainfed conditions, a maximum grain yield of 1600 kg ha⁻¹ was obtained. Under irrigation, N rates > 150 kg ha⁻¹ were required to produce grain yields of 2700 kg ha⁻¹ [35]. The relationship between water supply and fertilizer placement was also studied in the Central Highlands region [45]. At rates of 50 kg N ha⁻¹, maximum uptake by the crop was

52% and 44% of the applied N on the alluvial and tertiary soils, respectively. Without irrigation, maximum uptake decreased to 49% and 37% respectively with apparent losses of fertilizer N varying between 8% and 18%. Recovery of fertilizer N in the plant tops varied between 20% on the alluvial soil and 7% on the tertiary soil for broadcast and incorporated urea treatments, respectively. On the basaltic soil, banding of fertilizer N was the more effective method of application, resulting in a 52% recovery of N in the plant. This compared with a 24% recovery for broadcast-incorporated urea [45]. On the Darling Downs, grain yields of 5000–7000 kg ha⁻¹ were obtained under furrow irrigation when N application rates were in the range of 120–180 kg N ha⁻¹ [33].

Fertilizer N losses

Ammonia volatilization losses of N from beneath sorghum crops can be expected when urea is the major form of fertilizer applied. Laboratory studies on soils from the Central Highlands indicated recovery of urea was better from the tertiary soil (72–92%) than from the red earth (55–83%) [13]. The latter soil had only a small affinity for ammonium due to its high pH, low CEC and light texture, attributes which would contribute to greater ammonia volatilization and hence poor N recovery. Recovery of fertilizer N was increased by banding instead of mixing, with losses of fertilizer N reduced by 10%. This finding was confirmed in a field study where banding reduced losses by 20% [45]. Banding of urea into the soil at sowing is recommended for dryland sorghum production.

Leaching of added or mineralized N has been found to be a major pathway of N removal from soils used for sorghum production. Leached N accumulated at depth in the soil profile and amounted to 300 kg NO₃⁻ N ha⁻¹ when a vertisol was subjected to two years of exceptionally high rainfall. These losses could be minimized by crop rotation where sorghum is replaced one year in 10 with either pasture grass, green panic (*Panicum maximum* cv. Petrie) or by deeper rooting crops such as bulrush millet (*Pennisetum glaucum*) or sunflower (*Helianthus annuus*) [12]. This allows retrieval of the nitrate-N from the sub-soil, giving good productivity of pasture or alternative crop and reducing the N fertilizer requirement of the subsequent sorghum crop. There is the added benefit of reducing entry of nitrate into groundwater systems.

Denitrification has been found to be responsible for large losses of N at depth beneath cropping systems in

the subtropics [59]. Beneath a mung bean (*Vigna mungo*) crop on a heavy black earth, losses of 31 kg N ha⁻¹ yr⁻¹ were recorded although the loss was limited by the lack of a suitable energy source at depth. Comparable losses could be expected from beneath a sorghum crop growing on this soil as similar quantities of N (300 kg N ha⁻¹) were found to accumulate under both crops.

Cotton

Cotton is an important summer crop in irrigated areas of the sub-humid zones with 250,000 ha sown during the 1991–92 season [1]. The value of the crop in that year was about \$1 billion. Anhydrous ammonia is the preferred N fertilizer although both urea and ammonium nitrate (Nitram) are finding increasing use.

In the Central Highlands region, where 25% of the Queensland cotton crop is now produced, low yields of lint were obtained from the initial plantings. The need for large amounts of N fertilizer was soon demonstrated, and a base application rate of 120 kg N ha⁻¹ is now recommended [30]. Other experimentation in the area had shown that N rates of 112 kg N ha⁻¹ were sufficient during the early growth of cotton under irrigation, but much lower rates were sufficient if rates > 150 kg N ha⁻¹ were used on the previous crop [39]. A comparison of urea and nitram on the cotton crop showed no significant difference in lint yields [34]. In the Namoi Valley in the southern zone, 30% of the N applied to the crop as anhydrous ammonia was taken up by the crop 120 days after sowing [15]. Maximum uptake at this time was 108 kg N ha⁻¹ when 173 kg N ha⁻¹ was applied. The preceding crop also influenced the N requirements of irrigated cotton [32]. Without applied N, cotton yielded least after cotton or sorghum and most after wheat or fallow. Similarly, the response to fertilizer N was greatest after cotton or sorghum and least after wheat or fallow. Calculations using an N response curve showed reductions of fertilizer requirement of between 66 and 111 kg N ha⁻¹ when cotton was grown after soybeans, wheat and fallow compared with cotton after cotton. Cotton yield and N requirements are thus influenced by crop rotation and tillage method [16]. An evaluation of a cotton-wheat rotation with continuous cotton under maximum and minimum tillage showed that lint yield from the rotation was 1.3% and 7.1% higher than minimum and maximum tillage respectively. Nitrogen fertilizer uptake was also lower, averaging 145 kg ha⁻¹ for the rotation compared with 189 kg ha⁻¹ under minimum tillage and

210 kg ha⁻¹ for maximum disturbance for continuous cotton.

Recovery of fertilizer N

Apparent recovery of N by a cotton crop averages about 40% of the applied N and is seldom > 50% [15, 16]. Factors identified as affecting recovery include water stress, crop rotation, waterlogging and soil compaction [32], the latter two having considerable influence on the loss of fertilizer N through biological denitrification. Since a large quantity of fertilizer N is presumed lost via this process, the effectiveness of nitrification inhibitors in reducing denitrification losses and thereby increasing yields has been evaluated. Recovery of applied N increased from 57% in the absence of inhibitor to 78% and 92% in the presence of wax-coated calcium carbide and 2-ethynylpyridine respectively [27]. However, the prohibitive cost of using 2-ethynylpyridine in the field makes the use of coated calcium carbide the better alternative, as the cost of 30 kg of the latter equals the cost of 10 kg urea [27].

Micrometeorological techniques have been used to measure losses of applied N through ammonia volatilization [19]. Loss from anhydrous ammonia applied to a cotton crop at a rate of 107 kg N ha⁻¹ was found to be 1 kg N ha⁻¹ with maximum errors associated with this value being about 20%. In this study, the loss was reduced by the design of a suitable applicator and a mean injection depth of 12.4 cm.

Conclusions

The use of N fertilizer to maintain economic yields of sugarcane, cotton and grain crops is common practice in subtropical agriculture in Australia. Greater farm profitability, however, will come from using N fertilizer more efficiently. This requires an awareness of the processes that cause N loss and adjusting management practices to counteract them. The most common form of N fertilizer used in all four climatic zones in subtropical Australia is urea. If urea is not applied using recommended procedures, large losses of N can occur via ammonia volatilization, denitrification and leaching. For ammonia volatilization, these procedures could include: (a) placing the urea beneath the soil surface (b) applying adequate irrigation to wash the urea into the soil (particularly with green cane trash blanketing in sugarcane) and (c) using urease inhibitors to delay hydrolysis until rainfall occurs. For denitrifica-

tion, appropriate measures may include either delaying nitrate formation through the use of nitrification inhibitors or applying urea at reduced rates so that less nitrate is present in the soil profile. This latter procedure is important as it will reduce the quantity of nitrous oxide being emitted by increasing the conversion of nitrous oxide to dinitrogen, thus reducing trace gas contamination of the atmosphere. For leaching, the procedures are not well defined. This process occurs more widely in cropping systems than in pastures. The practice of using deeper rooting crops in rotation with the more traditional crops has been successful in retrieving some of the N from deep in the soil profile. This practice is recommended for soils where nitrate accumulates at depth beneath a cultivated crop over the growing season. Soil erosion can be a major avenue for N loss from cropping systems. Procedures such as green cane trash blanketing in sugar-cane and the elimination of stubble burning from grain crops significantly reduces the amount of N lost through erosion. The retention and/or the incorporation of organic matter in this way also increases soil fertility and improves soil structure.

Recent research with sugar-cane has indicated a reduction in N losses with an increase in N use efficiency. By synchronising both N and water application rates to the needs of the crop, a reduction in the quantity of nitrate entering groundwater and river systems may be expected. Further developments with nitrification inhibitors may result in less N use on grain crops since N is available to the plant for a longer period. Increased fertilizer recovery (or uptake efficiency) can therefore be expected for most crops providing this knowledge is imparted to and accepted by growers as a means of decreasing production costs.

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